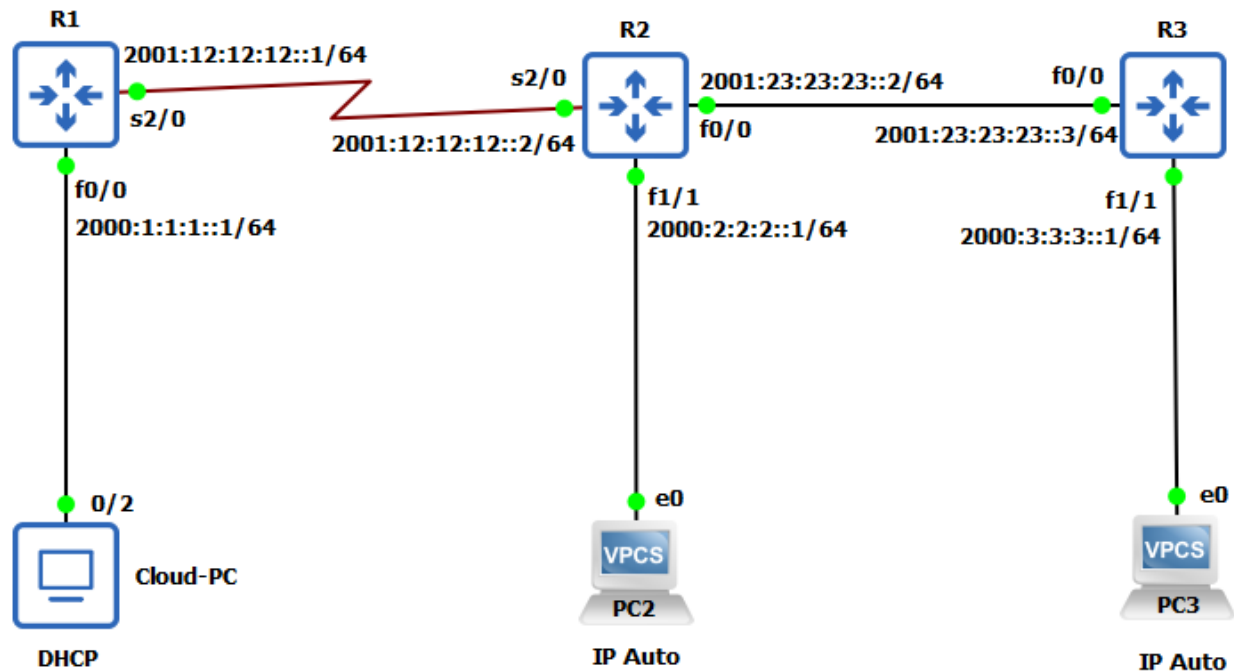


Static Routing – IPv6



Task

1. Configure routers with IPv6 address as shown in topology (Refer Lab-2)
2. Configure IPv6 static routes (next-hop / exit interface / link-local) on routers to provide network reachability.
3. Cloud-PC, PC2 & PC3 should be able to ping each other.

Note: Cloud-PC is a cloud which bridges the R1 router with loopback adapter on real PC.

Static routing on R1 Router

```
R1#config t
R1(config)#ipv6 route 2000:2:2:2::/64 2001:12:12:12::2
R1(config)#ipv6 route 2001:23:23:23::/64 2001:12:12:12::2
R1(config)#ipv6 route 2000:3:3:3::/64 2001:12:12:12::2
R1(config)#exit
R1#
```

Static routing on R2 Router

```
R2#config t
R2(config)#ipv6 route 2000:1:1:1::/64 2001:12:12:12::1
R2(config)#ipv6 route 2000:3:3:3::/64 2001:23:23:23::3
R2(config)#exit
R2#
```

Static routing on R3 Router

```

R3#config t
R3(config)#ipv6 route 2000:2:2:2::/64 2001:23:23:23::2
R3(config)#ipv6 route 2001:12:12:12::/64 2001:23:23:23::2
R3(config)#ipv6 route 2000:1:1:1::/64 2001:23:23:23::2
R3(config)#exit
R3#

```

✓ Verification & Testing:**R1#sh ip route static**

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

R1#

R1#sh ipv6 route

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
 B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
 H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
 IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
 ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
 O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP

C 2000:1:1:1::/64 [0/0]
 via FastEthernet0/0, directly connected

L 2000:1:1:1::1/128 [0/0]
 via FastEthernet0/0, receive

S 2000:2:2:2::/64 [1/0]
 via 2001:12:12:12::2

S 2000:3:3:3::/64 [1/0]
 via 2001:12:12:12::2

C 2001:12:12:12::/64 [0/0]
 via Serial2/0, directly connected

L 2001:12:12:12::1/128 [0/0]
 via Serial2/0, receive

S 2001:23:23:23::/64 [1/0]
 via 2001:12:12:12::2

L FF00::/8 [0/0]
 via Null0, receive

R1#

R2#sh ipv6 route static

IPv6 Routing Table - default - 9 entries

Codes: C - Connected, L - Local, **S - Static**, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S 2000:1:1:1::/64 [1/0]

via 2001:12:12:12::1

S 2000:3:3:3::/64 [1/0]

via 2001:23:23:23::3

R2#

R3#sh ipv6 route updated

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, **S - Static**, **U** - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S 2000:1:1:1::/64 [1/0]

via 2001:23:23:23::2

Last updated 16:28:40 07 October 2020**S 2000:2:2:2::/64 [1/0]**

via 2001:23:23:23::2

Last updated 16:28:40 07 October 2020**C 2000:3:3:3::/64 [0/0]**

via FastEthernet1/1, directly connected

Last updated 16:14:43 07 October 2020**L 2000:3:3:3::1/128 [0/0]**

via FastEthernet1/1, receive

Last updated 16:14:44 07 October 2020**S 2001:12:12:12::/64 [1/0]**

via 2001:23:23:23::2

Last updated 16:29:01 07 October 2020**C 2001:23:23:23::/64 [0/0]**

via FastEthernet0/0, directly connected

Last updated 16:14:43 07 October 2020**L 2001:23:23:23::3/128 [0/0]**

via FastEthernet0/0, receive

Last updated 16:14:44 07 October 2020**L FF00::/8 [0/0]**

via Null0, receive

Last updated 16:14:38 07 October 2020

R3#

R1#sh ipv6 route 2000:3:3:3::/64

Routing entry for 2000:3:3:3::/64

Known via "static", distance 1, metric 0

Route count is 1/1, share count 0

Routing paths:

2001:12:12:12::2

Last updated 00:05:33 ago

R2#sh ipv6 route 2000:1:1:1::/64

Routing entry for 2000:1:1:1::/64

Known via "static", distance 1, metric 0

Route count is 1/1, share count 0

Routing paths:

2001:12:12:12::1

Last updated 00:05:26 ago

R3#sh ipv6 route 2000:1:1:1::/64

Routing entry for 2000:1:1:1::/64

Known via "static", distance 1, metric 0

Route count is 1/1, share count 0

Routing paths:

2001:23:23:23::2

Last updated 00:05:33 ago

R1#sh ipv6 static

IPv6 Static routes Table - default

Codes: * - installed in RIB, u/m - Unicast/Multicast only

U - Per-user Static route

N - ND Static route

M - MIP Static route

P - DHCP-PD Static route

R - RHI Static route

* 2000:2:2:2::/64 via 2001:12:12:12::2, distance 1

* 2000:3:3:3::/64 via 2001:12:12:12::2, distance 1

* 2001:23:23:23::/64 via 2001:12:12:12::2, distance 1

R2#sh ipv6 static

IPv6 Static routes Table - default

Codes: * - installed in RIB, u/m - Unicast/Multicast only

U - Per-user Static route

N - ND Static route

M - MIP Static route

P - DHCP-PD Static route

R - RHI Static route

* 2000:1:1:1::/64 via 2001:12:12:12::1, distance 1

* 2000:3:3:3::/64 via 2001:23:23:23::3, distance 1

R3#sh ipv6 static

IPv6 Static routes Table - default

Codes: * - installed in RIB, u/m - Unicast/Multicast only

U - Per-user Static route

N - ND Static route
M - MIP Static route
P - DHCP-PD Static route
R - RHI Static route

- * 2000:1:1:1::/64 via 2001:23:23:23::2, distance 1
- * 2000:2:2:2::/64 via 2001:23:23:23::2, distance 1
- * 2001:12:12:12::/64 via 2001:23:23:23::2, distance 1

Ping PC2 & PC3 from R1

R1#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/48/64 ms

R1#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 64/69/88 ms

Ping Cloud-PC & PC3 from R2

R2#ping ipv6 2000:1:1:1:d5d6:56d2:75d0:b647

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:D5D6:56D2:75D0:B647, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/48/72 ms

R2#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/49/76 ms

Ping Cloud-PC & PC2 from R3

R3#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/50/80 ms

R3#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/45/52 ms

Ping PC2 & PC3 from Cloud-PC

C:\Users\Ratan>ping 2000:2:2:2050:79ff:fe66:6801

Pinging 2000:2:2:2:2050:79ff:fe66:6801 with 32 bytes of data:

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=53ms

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=34ms

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=38ms

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=38ms

Ping statistics for 2000:2:2:2:2050:79ff:fe66:6801:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 34ms, Maximum = 53ms, Average = 40ms

C:\Users\Ratan>ping 2000:3:3:3:2050:79ff:fe66:6800

Pinging 2000:3:3:3:2050:79ff:fe66:6800 with 32 bytes of data:

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=54ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=59ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=56ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=55ms

Ping statistics for 2000:3:3:3:2050:79ff:fe66:6800:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 54ms, Maximum = 59ms, Average = 56ms

Ping Cloud-PC & PC3 from PC2

PC2> ping 2000:3:3:3:2050:79ff:fe66:6800/64

2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=1 ttl=60 time=42.342 ms

2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=2 ttl=60 time=43.229 ms

2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=3 ttl=60 time=43.354 ms

2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=4 ttl=60 time=43.327 ms

2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=5 ttl=60 time=42.294 ms

PC2> ping 2000:1:1:1:d5d6:56d2:75d0:b647

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=1 ttl=62 time=37.384 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=2 ttl=62 time=42.307 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=3 ttl=62 time=42.295 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=4 ttl=62 time=41.384 ms

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=5 ttl=62 time=42.225 ms

Ping Cloud-PC & PC2 from PC3

PC3> ping 2000:2:2:2:2050:79ff:fe66:6801

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=1 ttl=60 time=71.845 ms

```
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=2 ttl=60 time=42.384 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=3 ttl=60 time=43.036 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=4 ttl=60 time=43.117 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=5 ttl=60 time=42.182 ms
```

PC3> ping 2000:1:1:1:d5d6:56d2:75d0:b647

```
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=1 ttl=61 time=72.007 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=2 ttl=61 time=61.952 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=3 ttl=61 time=62.464 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=4 ttl=61 time=64.045 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=5 ttl=61 time=64.138 ms
```

Remove the previous configured static route and reconfigure the static route with self-exit interface

Static routing on R1 Router

```
R1#config t
R1(config)#no ipv6 route 2000:2:2:2::/64 2001:12:12:12::2
R1(config)#no ipv6 route 2001:23:23:23::/64 2001:12:12:12::2
R1(config)#no ipv6 route 2000:3:3:3::/64 2001:12:12:12::2
R1(config)#ipv6 route 2000:2:2:2::/64 s2/0
R1(config)#ipv6 route 2001:23:23:23::/64 s2/0
R1(config)#ipv6 route 2000:3:3:3::/64 s2/0
R1(config)#exit
R1#
```

Static routing on R2 Router

```
R2#config t
R2(config)#no ipv6 route 2000:1:1:1::/64 2001:12:12:12::1
R2(config)#no ipv6 route 2000:3:3:3::/64 2001:23:23:23::3
R2(config)#ipv6 route 2000:1:1:1::/64 s2/0
R2(config)#ipv6 route 2000:3:3:3::/64 fa0/0
R2(config)#exit
R2#
```

Static routing on R3 Router

```
R3#config t
R3(config)#no ipv6 route 2000:2:2:2::/64 2001:23:23:23::2
R3(config)#no ipv6 route 2001:12:12:12::/64 2001:23:23:23::2
R3(config)#no ipv6 route 2000:1:1:1::/64 2001:23:23:23::2
R3(config)#ipv6 route 2000:2:2:2::/64 fa0/0
R3(config)#ipv6 route 2001:12:12:12::/64 fa0/0
R3(config)#ipv6 route 2000:1:1:1::/64 fa0/0
R3(config)#exit
R3#
```

✓ **Verification & Testing:****R1#sh ipv6 route static**

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, **S - Static**, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S 2000:2:2:2::/64 [1/0]

via Serial2/0, directly connected

S 2000:3:3:3::/64 [1/0]

via Serial2/0, directly connected

S 2001:23:23:23::/64 [1/0]

via Serial2/0, directly connected

R2#sh ipv6 route static

IPv6 Routing Table - default - 9 entries

Codes: C - Connected, L - Local, **S - Static**, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S 2000:1:1:1::/64 [1/0]

via Serial2/0, directly connected

S 2000:3:3:3::/64 [1/0]

via FastEthernet0/0, directly connected

R3#sh ipv6 route static

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, **S - Static**, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S 2000:1:1:1::/64 [1/0]

via FastEthernet0/0, directly connected

S 2000:2:2:2::/64 [1/0]

via FastEthernet0/0, directly connected

S 2001:12:12:12::/64 [1/0]

via FastEthernet0/0, directly connected

R1#sh ipv6 static

IPv6 Static routes Table - default

Codes: * - installed in RIB, u/m - Unicast/Multicast only

U - Per-user Static route
N - ND Static route
M - MIP Static route
P - DHCP-PD Static route
R - RHI Static route

- * 2000:2:2:2::/64 via Serial2/0, distance 1
- * 2000:3:3:3::/64 via Serial2/0, distance 1
- * 2001:23:23:23::/64 via Serial2/0, distance 1

R2#sh ipv6 static

IPv6 Static routes Table - default

Codes: * - installed in RIB, u/m - Unicast/Multicast only

U - Per-user Static route
N - ND Static route
M - MIP Static route
P - DHCP-PD Static route
R - RHI Static route

- * 2000:1:1:1::/64 via Serial2/0, distance 1
- * 2000:3:3:3::/64 via FastEthernet0/0, distance 1

R3#sh ipv6 static

IPv6 Static routes Table - default

Codes: * - installed in RIB, u/m - Unicast/Multicast only

U - Per-user Static route
N - ND Static route
M - MIP Static route
P - DHCP-PD Static route
R - RHI Static route

- * 2000:1:1:1::/64 via FastEthernet0/0, distance 1
- * 2000:2:2:2::/64 via FastEthernet0/0, distance 1
- * 2001:12:12:12::/64 via FastEthernet0/0, distance 1

Ping PC2 & PC3 from R1

R1#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/62/72 ms

R1#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:
.....

Success rate is 0 percent (0/5)

Ping Cloud-PC & PC3 from R2

R2#ping ipv6 2000:1:1:1:d5d6:56d2:75d0:b647

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:D5D6:56D2:75D0:B647, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/44/48 ms

R2#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

Ping Cloud-PC & PC2 from R3

R3#ping ipv6 2000:1:1:1:d5d6:56d2:75d0:b647

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:D5D6:56D2:75D0:B647, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

R3#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

Imp Note: In IPv6, exit interface option for static route is only applicable for point-to-point links (serial interface) and not for multi-access links (FastEthernet interface). So, ping response will fail for traffic exiting from fastethernet interfaces.

Remove the previous configured static route and reconfigure the static route with exit interface and link-local address.

First, we need to know the link-local address on each router.

R1#sh ipv6 interface s2/0 | include link-local

IPv6 is enabled, link-local address is FE80::C801:20FF:FE88:8

No Virtual link-local address(es):

R2#sh ipv6 interface s2/0 | include link-local

IPv6 is enabled, link-local address is FE80::C802:CFF:FE24:8

No Virtual link-local address(es):

R2#sh ipv6 interface fa0/0 | include link-local

IPv6 is enabled, link-local address is FE80::C802:CFF:FE24:8

No Virtual link-local address(es):

R3#sh ipv6 interface fa0/0 | include link-local

IPv6 is enabled, link-local address is FE80::C803:1EFF:FE00:8

No Virtual link-local address(es):

Static routing on R1 Router

```

R1#config t
R1(config)#no ipv6 route 2000:2:2:2::/64 s2/0
R1(config)#no ipv6 route 2001:23:23:23::/64 s2/0
R1(config)#no ipv6 route 2000:3:3:3::/64 s2/0
R1(config)#ipv6 route 2000:2:2:2::/64 s2/0 FE80::C802:CFF:FE24:8
R1(config)#ipv6 route 2001:23:23:23::/64 s2/0 FE80::C802:CFF:FE24:8
R1(config)#ipv6 route 2000:3:3:3::/64 s2/0 FE80::C802:CFF:FE24:8
R1(config)#exit
R1#

```

Static routing on R1 Router

```

R2#config t
R2(config)#no ipv6 route 2000:1:1:1::/64 s2/0
R2(config)#no ipv6 route 2000:3:3:3::/64 fa0/0
R2(config)#ipv6 route 2000:1:1:1::/64 s2/0 FE80::C801:20FF:FE88:8
R2(config)#ipv6 route 2000:3:3:3::/64 fa0/0 FE80::C803:1EFF:FE00:8
R2(config)#exit
R2#

```

Static routing on R1 Router

```

R3#config t
R3(config)#no ipv6 route 2000:2:2:2::/64 fa0/0
R3(config)#no ipv6 route 2001:12:12:12::/64 fa0/0
R3(config)#no ipv6 route 2000:1:1:1::/64 fa0/0
R3(config)#ipv6 route 2000:2:2:2::/64 fa0/0 FE80::C802:CFF:FE24:8
R3(config)#ipv6 route 2001:12:12:12::/64 fa0/0 FE80::C802:CFF:FE24:8
R3(config)#ipv6 route 2000:1:1:1::/64 fa0/0 FE80::C802:CFF:FE24:8
R3(config)#exit
R3#

```

✓ Verification & Testing:**R1# sh ipv6 route static**

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

```

S 2000:2:2:2::/64 [1/0]
  via FE80::C802:CFF:FE24:8, Serial2/0
S 2000:3:3:3::/64 [1/0]
  via FE80::C802:CFF:FE24:8, Serial2/0
S 2001:23:23:23::/64 [1/0]
  via FE80::C802:CFF:FE24:8, Serial2/0

```

R2#sh ipv6 route static

IPv6 Routing Table - default - 9 entries

Codes: C - Connected, L - Local, **S - Static**, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S 2000:1:1:1::/64 [1/0]

via FE80::C801:20FF:FE88:8, Serial2/0

S 2000:3:3:3::/64 [1/0]

via FE80::C803:1EFF:FE00:8, FastEthernet0/0

R3#sh ipv6 route static

IPv6 Routing Table - default - 8 entries

Codes: C - Connected, L - Local, **S - Static**, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

S 2000:1:1:1::/64 [1/0]

via FE80::C802:CFF:FE24:8, FastEthernet0/0

S 2000:2:2:2::/64 [1/0]

via FE80::C802:CFF:FE24:8, FastEthernet0/0

S 2001:12:12:12::/64 [1/0]

via FE80::C802:CFF:FE24:8, FastEthernet0/0

Ping PC2 & PC3 from R1

R1#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/51/64 ms

R1#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/60/64 ms**Ping Cloud-PC & PC3 from R2**

R2#ping ipv6 2000:1:1:1:d5d6:56d2:75d0:b647

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:D5D6:56D2:75D0:B647, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/43/48 ms

R2#ping ipv6 2000:3:3:3:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:3:3:3:2050:79FF:FE66:6800, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/50/60 ms

Ping Cloud-PC & PC2 from R3

R3#ping ipv6 2000:1:1:1:d5d6:56d2:75d0:b647

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:D5D6:56D2:75D0:B647, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 60/65/68 ms

R3#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 24/35/64 ms

Ping PC2 & PC3 from Cloud-PC

C:\Users\Ratan>ping 2000:2:2:2:2050:79ff:fe66:6801

Pinging 2000:2:2:2:2050:79ff:fe66:6801 with 32 bytes of data:

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=51ms

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=41ms

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=37ms

Reply from 2000:2:2:2:2050:79ff:fe66:6801: time=41ms

Ping statistics for 2000:2:2:2:2050:79ff:fe66:6801:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 37ms, Maximum = 51ms, Average = 42ms

C:\Users\Ratan>ping 2000:3:3:3:2050:79ff:fe66:6800

Pinging 2000:3:3:3:2050:79ff:fe66:6800 with 32 bytes of data:

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=62ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=58ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=56ms

Reply from 2000:3:3:3:2050:79ff:fe66:6800: time=65ms

Ping statistics for 2000:3:3:3:2050:79ff:fe66:6800:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 56ms, Maximum = 65ms, Average = 60ms

Ping Cloud-PC & PC3 from PC2**PC2> ping 2000:1:1:1:d5d6:56d2:75d0:b647**

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=1 ttl=62 time=40.104 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=2 ttl=62 time=41.137 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=3 ttl=62 time=42.150 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=4 ttl=62 time=41.144 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=5 ttl=62 time=42.320 ms

PC2> ping 2000:3:3:3:2050:79ff:fe66:6800

2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=1 ttl=60 time=45.241 ms
2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=2 ttl=60 time=41.132 ms
2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=3 ttl=60 time=42.164 ms
2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=4 ttl=60 time=41.116 ms
2000:3:3:3:2050:79ff:fe66:6800 icmp6_seq=5 ttl=60 time=41.190 ms

Ping Cloud-PC & PC2 from PC3**PC3> ping 2000:1:1:1:d5d6:56d2:75d0:b647**

2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=1 ttl=61 time=74.001 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=2 ttl=61 time=64.983 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=3 ttl=61 time=64.003 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=4 ttl=61 time=85.091 ms
2000:1:1:1:d5d6:56d2:75d0:b647 icmp6_seq=5 ttl=61 time=64.143 ms

PC3> ping 2000:2:2:2:2050:79ff:fe66:6801

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=1 ttl=60 time=41.465 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=2 ttl=60 time=42.037 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=3 ttl=60 time=41.105 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=4 ttl=60 time=41.250 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=5 ttl=60 time=41.152 ms

Important Commands:

```
sh ipv6 route  
sh ipv6 route static  
sh ipv6 route updated  
sh ipv6 static
```

Ratan